Introduction to Ab Initio Nuclear Theory

Lecture 4: **Quantum Monte Carlo**

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Contents

"They had devised a new way to use computers beyond simply performing accelerated calculations that humans could already do."
Sophia Chen, APS news article on A. Rosebluth, a pioneer of the Metropolis-Hastings algorithm

1 The Quantum Monte Carlo Family of approaches

Quantum Monte Carlo refers to a family of many-body methods that solves the many-body Schröndiger's equation stochastically. The most well-known methods for nuclear theory are Variational Monte Carlo (VMC), Diffusion Monte Carlo (DMC), Auxiliary Field Diffusion Monte Carlo (AFDMC), and Greens Functions Monte Carlo (GFMC). We will discuss the first two, VMC and DMC, in detail. The last two AFDMC and GFMC are generalization of DMC to spin-dependent interactions and follow the same principles as DMC. Note DMC and GFMC were first developed to study the many-electron ground state.

2 The Monte Carlo quadrature

2.1 Monte Carlo Integration

Many concepts discussed in this Lecture can be understood purely from a numerical analysis perspective: we are integrating Schröndiger's equation. To facilitate the discussion we will start from the essential elements of the Monte Carlo quadrature.

Monte Carlo relies on estimating integrals by interpreting them as expectations of statistical distributions; so let's start from there. We'll start from the case of a single variable and generalize to many variables, where Monte Carlo integration has its edge. The expectation of a function f of a continuous random variable X distributed according to a probability distribution $p(x)$.

$$\langle f(X)\rangle = \int_{-\infty}^{\infty} dx\, p(x) f(x) \tag{1}$$

we will take $p(x) = w(x)/(b-a)$ for $x \in [a,b]$ and zero elsewhere; this is sufficient for most practical purposes since even when the integration extends to infinity, often in practice one integrates only up to some large number. This brings the expectation to the form

$$I = \frac{1}{b-a}\int_a^b dx\, w(x) f(x) = \langle f(X)\rangle \tag{2}$$

This is of the form of a general Gaussian quadrature where the integral is subsequently descritized on a set of points that correspond to extrema or nodes of a family of orthogonal polynomials. Here we'll take an alternative approach and by interpreting $w(x)$ as a probability density we will evaluate the integral stochastically. Taking the definition of a variance of a random variable we can also define

$$\text{var}[f(X)] = \left\langle [f(x) - \langle f(X)\rangle]^2 \right\rangle = \left\langle f(X)^2 \right\rangle - \langle f(X)\rangle^2 \tag{3}$$

$$= \frac{1}{b-a}\int_b^a dx\, w(x) f^2(x) - \left(\frac{1}{b-a}\int_a^b dx\, w(x) f(x)\right)^2 \tag{4}$$

These two quantities, $\langle f(X)\rangle$ and $\text{var}[f(X)]$, are named *population mean* and *population variance*, respectively, and thy are the ones we wish to calculate. The former will be the value of the integral we are after and the latter will define the error of our estimation via the population standard deviation $\sqrt{\text{var}[f(X)]}$. To estimate these quantities we will define what is usually referred to as *estimators*, that is, recipes for estimating a statistical property of a population using observed data; in our case the observed data will be evaluations of the function $f(x)$ for a sample of values for x.

Assume a sample of n random variables $X_1, X_2, \ldots, X_n$ are drawn from $w(x)$. Applying the function f to each one of them leads to a new sample of random variables $f(X_1), f(X_2), \ldots, f(X_n)$. We call the *sample mean* the mean of the second sample

$$\bar{f} = \frac{1}{n}\sum_{i=1}^{n} f(X_i) \ . \tag{5}$$

Making a distinction as to what is considered a random variable here is important. The variables X_i are random variables and so are $f(X_i)$ since the function of a random variable is also a random variable (albeit following a different distribution). Consequently $\bar{f}$ is also a random variable as it is the sum of the random variables $f(X_i)$, the sum being, of course, a function. Since $\bar{f}$ is a random variable we can calculate its expectation

$$\langle \bar{f} \rangle = \left\langle \frac{1}{n}\sum_{i=1}^{n} f(X_i) \right\rangle = \frac{1}{n}\sum_{i=1}^{n} \langle f(X_i) \rangle = \langle f(X) \rangle \ . \tag{6}$$

Note that since the sample mean is computed by a finite number of samples n, $\bar{f}$ itself could be quite far from $\langle f(X) \rangle$ (e.g., consider the case $n = 2$). However, Eq. (6) means that the *expectation* of the sample mean is equal to the population mean. Even better, one can show that as n grows larger the probability of $\bar{f}$ being equal to $\langle f(X) \rangle$ increases. This motivates us to choose $\bar{f}$ as an estimator for $\langle f(X) \rangle$, and consequently as an approximation to our integral. How good of an approximation $\bar{f}$ is for $\langle f(X) \rangle$ or our integral is related to the variance of the sample mean (again, remember, the sample mean is a random variable):

$$\mathrm{var}[\bar{f}] = \mathrm{var}\left[\frac{1}{n}\sum_{i=1}^{n} f(X_i)\right] = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{var}[f(X)] = \frac{1}{n}\mathrm{var}[f(X)] \ , \tag{7}$$

where we have used the identity for the variance of a sum of random variables. Hence the standard deviation of the sample mean tells of the error made the sample mean as an estimator, is

$$\sqrt{\mathrm{var}[\bar{f}]} = \frac{1}{\sqrt{n}}\sqrt{\mathrm{var}[f(X)]} \ . \tag{8}$$

This result is at the heart of Monte Carlo quadrature's success in evaluating multidimensional integrals. The variance drops as $1/\sqrt{n}$ and that scaling doesn't depend on the space's dimensionality. Hence the same efficiency is attained in higher dimensional integrals as well, beating all other methods; more on this later.

We'll now play the same game with $\mathrm{var}[f(X)]$ and find an estimator for it. Intuitively, we turn to the sample variance

$$e_{\mathrm{var}} = \bar{f^2} - (\bar{f})^2 = \frac{1}{n}\sum_{i=1}^{n} f^2(X_i) - \left[\frac{1}{n}\sum_{i=1}^{n} f(X_i)\right]^2 \ . \tag{9}$$

Again, being explicit: e_{var} is a function of random variable and hence a random variable itself. its expectation is

$$\langle e_{\mathrm{var}} \rangle = \frac{n-1}{n}\mathrm{var}[f(X)] \ . \tag{10}$$

So we can use the sample variance (e_{var}) to define an estimator for the population variance ($\mathrm{var}[f(X)]$) which in turn will allow us to evaluate the error made by the sample mean ($\bar{f}$) as an estimator for population mean ($\langle f(X) \rangle$), which is our integral. Putting everything together the Monte Carlo estimation for our integral is

$$\int_a^b dx\, w(x)f(x) \approx (b-a)\bar{f} \pm (b-a)\sqrt{\mathrm{var}[\bar{f}]} \tag{11}$$

$$\approx \frac{(b-a)}{n}\sum_{i=1}^{n} f(X_i) \pm \frac{b-a}{\sqrt{n-1}}\sqrt{\frac{1}{n}\sum_{i=1}^{n} f^2(X_i) - \left[\frac{1}{n}\sum_{i}^{n} f(X_i)\right]^2}\ , \tag{12}$$

where we employed the central limit theorem that allows us to treat $\bar{f}$ as a Gaussian distributed variable for large n, regardless of what $w(x)$ was used to draw the samples of X_i. The generalization for d-dimensions is straightforward and results in promoting $(b-a)$ to a d-dimensional volume V leading to the prescription

$$\int_V d^d x\, w(x)f(\mathbf{x}) \approx V\bar{f} \pm V\sqrt{\mathrm{var}[\bar{f}]} \tag{13}$$

$$\approx \frac{V}{n}\sum_{i=1}^{n} f(X_i) \pm \frac{V}{\sqrt{n-1}}\sqrt{\frac{1}{n}\sum_{i=1}^{n} f^2(X_i) - \left[\frac{1}{n}\sum_{i}^{n} f(X_i)\right]^2} \tag{14}$$

2.2 Importance sampling

Importance sampling provides a more efficient way of sampling for Monte Carlo integration. In essence, it requires that we come up with a function that should be qualitatively similar to $f(x)$ and use that to guide the sampling. To see how this is done let's first see a closely related method which is the *inverse transform sampling.* Consider the same integral we have been concerned with all this time

$$I = \int_a^b w(x)f(x)dx \tag{15}$$

It might have been your intuition all along to perform a change of variables to absorb the weight function in the integration measure. Let's do that:

$$I = \int_a^b w(x)f(x)dx = \int_0^{b-a} du f[x(u)]\ , \tag{16}$$

where

$$u(x) = \int_a^x w(x')dx \tag{17}$$

and $x(u)$ is the inverse of the (hopefully invertible) $u(x)$. Now if we were to evaluate this integral with Monte Carlo quadrature, we would employ Eq. (12) for the function of u, $f(x(u))$ and sample random variables U_i' from a uniform distribution (i.e., there is no $w(x)$).

Now, along the same lines, we can use this technique to improve our sampling for any integral. Indeed, consider the integral of a function $f(x)$ that is similar to a more well-behaved one $w(x)$ (we'll explain what well-behaved means here in a bit). Then we might want to do the following:

$$\int_a^b dx f(x) = \int_a^b dx w(x)\frac{f(x)}{w(x)} = \int_0^{b-a} du \frac{f[x(u)]}{w[x(u)]} = \frac{b-a}{n}\sum_{i=1}^{n}\frac{f[X(U_i)]}{w[X(U_i)]}\ . \tag{18}$$

where we have made the extra step this time of providing the final Monte Carlo form where to be explicit the random numbers U_i are sampled from a uniform distribution and the random numbers $X(U_i)$ result from applying $u(x)$'s inverse to the these random numbers. If $w(x)$ has the general features of $f(x)$, i.e., similar asymptotic behaviour, similarly peaked, etc., then the random numbers $X(U_i)$ which will be distributed according to $w(x)$ will explore these regions more than others and we will waste no labor exploring regions where, e.g., $f(x)$ is tiny. In other words, the closer w is to f, the closer f/w is to 1 and the integral's variance decreases.

2.3 Metropolis-Hastings algorithm

Unfortunately this method breaks down as one moves to higher dimensions where performing a change of variables and inverting functions gets very complicated. To the rescue comes the Metropolis-Hastings algorithm which provides a sampling recipe that is guaranteed to eventually provide random numbers from a distribution of our choosing. Let's start by laying down the concept of a Markov Chain.

Up to this point we have been referring to sampling where each successive sample $X_1, X_2, \ldots$ is independent from each other and just drawn from a common distribution, e.g., $w(x)$. Now consider a case where the sample X_i depends solely on the value of sample X_{i-1}, that is, X_i and X_{i-1} are not independent any more, but they don't necessarily stem from the same probability distribution. This is known as a *random walk* and the resulting set of samples a *Markov Chain*. The extra freedom provided by abolishing the common distribution and producing each sample from the previous one according to some rule means that one can tune the rule such that the chain converges to a distribution. This is a non-trivial statement: a Markov chain, defined as a set of successively *dependent* samples can reach a state where the samples appear *independent* and drawn from a common distribution. If that reminds you the concept of equilibrium in statistical physics, you're right, and we'll get to that soon.

Back to our integral, we wanted a better way of inverting multidimensional functions so that we could produce random numbers according to a desired distribution. Given the convenient properties of Markov Chains, we can forget about inverting functions and use Markov Chains to produce samples distributed according to the desired distribution. This is Markov Chain Monte Carlo (MCMC) and the Metropolis Hastings algorithm is a special way of implementing it.

As alluded to above, the problem at hand can benefit from analogies from statistical physics. There a sufficient conditions for evolving towards equilibrium is the principle of detailed balance

$$w(X)T(X \to Y) = w(Y)T(Y \to X) \ , \tag{19}$$

where $T(X \to Y)$ is the rule that defines the Markov Chain, that is, the conditional probability that we will get sample Y in the next step of the chain if we now have sample X and $w(X)$ is the probability of starting the walk at X. Combined, $w(X)T(X \to Y)$ is the probability of starting at X and moving to Y and the detailed balance condition requires that it's equal to the probability of starting at Y and moving to X. A system that obeys detailed balance is at equilibrium: we have the condition, we'll see below how to achieve it.

For moving from one sample to the next in a Markov Chain one only needs the rule and doesn't need to be concerned with the probability distribution that each successive sample follows. We have mentioned in passing that defining dependent samples that are not drawn from a single common distribution means that in principle these samples follow different probability distributions, defining a sequence of probability distributions along with the samples. Hence, to see how we can arrive to the situation where each sample follows the same distribution we'll have to see how the probability distribution of the samples evolves as we step through the chain and converges to said single distribution. If $p_i(X)$ is the probability distribution followed by sample X_i at step i, then the next probability distribution, at step $i+1$ is

$$p_{i+1}(X) = p_i(X) + \int dY \left[p_i(Y)T(Y \to X) - p_i(X)T(X \to Y)\right] \ , \tag{20}$$

or in words, the probability of being at X is equal to the probability of coming to X minus the probability of leaving from X. You can clearly see now that once we reach a distribution $w(x)$ that follows detailed balance, the integrand of the equation above we'll vanish and we'll forever stay at $w(x)$ and the Markov-Chain converges. This is not enough to prove that the Markov-Chain will actually converge: if no sample follows $w(x)$, that is, if the random walk never visits $w(x)$ we'll never converge. However, one can show that it's enough to be in the region of $w(X)$, that is at any $p(X)/w(X)$ close to 1 and the Markov-Chain will evolve towards $w(X)$.

We have hence proved that a Markov-Chain will converge to $w(X)$ if it obeys detailed balance. But how do we reach detailed balance? Or in other words, given $w(X)$, what is the rule $T(X \to Y)$ that will guarantee that we'll eventually reach equilibrium ? The Metropolis-Hastings algorithm is the standard answer to this question. According to the Metropolis-Hastings algorithm we only have to split the rule into two: a rule for proposing a step $X \to Y$ and a rule for accepting it with the proposing rule is left to us to tune for efficiency. Then we compare the probability of that proposal for the step $X \to Y$ ever coming up if X was distributed according to $w(X)$ with the corresponding probability of the inverse proposal $Y \to X$ if Y was distributed according to $w(y)$ (note this is different from the proposal probability since for a proposal for step $X \to Y$ to ever come up one will already have to be at X). If the probability of proposal of the reverse step $Y \to X$ is larger, the step is accepted: we are moving in a direction that is very probable to come back from. If the probability of proposal of the reverse step $Y \to X$ is smaller, the step is accepted according to how smaller it is: if we're moving to a region where it's very hard to come back from we probably shouldn't go. Putting all of this in equations, the Metropolis-Hastings algorithm says

$$T(X \to Y) = \pi(X \to Y)\alpha(X \to Y) \tag{21}$$

where $\pi(X \to Y)$ is the proposal probability (left to us to choose) and $\alpha(X \to Y)$ is the acceptance probability equal to

$$\alpha(X \to Y) = \min[1, R(X \to Y)] \ . \tag{22}$$

where $(X \to Y)$ is the Metropolis-Hastings ratio,

$$R(X \to Y) = \frac{w(X)\pi(X \to Y)}{w(Y)\pi(Y \to X)} \ . \tag{23}$$

It is straightforward now to show that $T(X \to Y)$ as prescribed by the Metropolis-Hastings algorithm obeys detailed balance (actually, the acceptance probability $\alpha(X \to Y)$ was chosen so that it does.) Our problem is solved: given an integral of a function f that is similar to another function w, we don't need to perform any change of variables but simply construct a Markov-Chain of samples using the Metropolis-Hastings algorithm employing $w(x)$ as the target distribution and we'll achieve importance sampling.

3 Variational Monte Carlo (VMC)

3.1 Introduction

Variational Monte Carlo (VMC) is typically formulated in coordinate space. We will first set up the appropriate language to describe systems in coordinate space, and use it to express the Hamiltonian. Then we will discuss the concept of a trial wavefunction, that is, our best guess for the Hamiltonian's ground state. Finally we will reformulate Schröndiger's equation in a variational form.

3.2 The QMC-*esque* coordinate space language

A system of N particles in d dimensions is described using Nd-dimensional vector of all positions of all particles

$$\mathbf{X} = (r_{11}, r_{12}, \ldots, r_{1d}, \ldots r_{N1}, r_{N2}, \ldots, r_{Nd})^T \ , \tag{24}$$

where r_{ij} is the j-th coordinate of the i-th particle, e.g., the second (i.e., y coordinate) of the 6th particle is r_{62}. For example, for 2 particles in 3 dimensions ($N = 2,\ d = 3$),

$$\mathbf{X} = (r_{11}, r_{12}, r_{13}, r_{21}, r_{22}, r_{23},)^T \quad . \tag{25}$$

We are after the *ground state energy*of the system, E_0. This is usually the point of departure for any other observable. Variational Monte Carlo provides an upper limit to E_0.

3.3 The Hamiltonian

The many-body Hamiltonian can be written in coordinate space

$$\hat{H} = \sum_{i=1}^{N} \hat{K}_i + \sum_{i<j} \hat{V}_{ij}^{(2)} + \sum_{i<j<k=1}^{N} \hat{V}_{ijk}^{(3)} + \cdots \tag{26}$$

$$\hat{H} = \sum_{i=1}^{N} \left(-\frac{\hbar^2}{2m} \boldsymbol{\nabla}_i^2 \right) + \sum_{i<j=1}^{N} V^{(2)} \left(\mathbf{r}_i, \mathbf{r}_j \right) + \sum_{i<j<k=1}^{N} V^{(3)} \left(\mathbf{r}_i, \mathbf{r}_j\, \mathbf{r}_k \right) + \cdots \tag{27}$$

where $\boldsymbol{\nabla}_i$ is the vector $(\partial_{r_{i1}}, \partial_{r_{i2}}, \ldots, \partial_{r_{id}})$ acting on the coordinates of particle i. The superscript of the operators $V^{(n)}$ signifies an n-body interaction: in QMC methods dealing with $N > 2$-body interactions is relatively straightforward, which is a relief for the study of nuclear systems (see Lecture 3).

The form of the Hamiltonian, e.g., in Eq. (**??**), guides the choice for a trial wavefunction. Inspired by ideas from perturbation theory, we could write the Hamiltonian as $\hat{H} = \hat{H}_0 + \hat{H}_1$, where $\hat{H}_0$'s ground state can be easily determined, and use that as a trial wave-function. Alternatively, we could use a phenomenological approximation to the ground state of $\hat{H}$ for a trial wavefuntion. More details, below.

Finally, to pose the problem well, we need to specify the boundary conditions that are applied to the particles. Note that even though we will not be solving and differential equations directly, the boundary conditions will enter our calculations.

3.4 Variational Monte Carlo

We saw in Lecture 1 how the expectation value of the Hamiltonian with any state provides an upper bound for the ground state energy. There we also admitted that the simplicity of the expressions hides the complexity of calculating the corresponding integrals for a many-body system. We will address this now: one can evaluate these integrals with Monte Carlo quadrature taking advantage of its good scaling. Further, parametrizing the trial wavefunction by some set of parameters one turns the problem of finding the ground state into a multi-dimensional variational problem.

Variational Monte Carlo,given a trial wavefunctions $|\Psi_T\rangle$, evaluates

$$\begin{aligned}
E_V(\boldsymbol{\alpha}) &= \frac{\langle\Psi_T(\boldsymbol{\alpha})|\, H \,|\Psi_T(\boldsymbol{\alpha})\rangle}{\langle\Psi_T(\boldsymbol{\alpha})|\Psi_T(\boldsymbol{\alpha})\rangle} = \frac{\int d\mathbf{X} \Psi_T^{\star}\left(\mathbf{X};\boldsymbol{\alpha}\right) H \Psi_T\left(\mathbf{X};\boldsymbol{\alpha}\right)}{\int d\mathbf{X} \left|\Psi_T\left(\mathbf{X};\boldsymbol{\alpha}\right)\right|^2} = \frac{\int d\mathbf{X} |\Psi_T\left(\mathbf{X};\boldsymbol{\alpha}\right)|^2 \Psi_T^{-1}\left(\mathbf{X};\boldsymbol{\alpha}\right) H \Psi_T\left(\mathbf{X};\boldsymbol{\alpha}\right)}{\int d^d r \left|\Psi_T\left(\mathbf{X};\boldsymbol{\alpha}\right)\right|^2} \\
&= \frac{\int d\mathbf{X} |\Psi_T\left(\mathbf{X};\boldsymbol{\alpha}\right)|^2 \Psi_T^{-1}\left(\mathbf{X};\boldsymbol{\alpha}\right) H \Psi_T\left(\mathbf{X};\boldsymbol{\alpha}\right)}{\int d\mathbf{X} \left|\Psi_T\left(\mathbf{X};\boldsymbol{\alpha}\right)\right|^2} \\
&= \int d\mathbf{X} P_T(\mathbf{X}) E_L(\mathbf{X})
\end{aligned} \tag{28}$$

where the local energy and the configuration space trial probability density are

$$E_L(\mathbf{X};\boldsymbol{\alpha}) = \frac{H\Psi(\mathbf{X};\alpha)}{\Psi(\mathbf{X};\boldsymbol{\alpha})} \;, \quad P_T(\mathbf{X};\alpha) = \frac{|\Psi(\mathbf{X};\alpha)|^2}{\int d\mathbf{X}|\Psi(\mathbf{X};\boldsymbol{\alpha})|^2} \tag{29}$$

Given our discussion on Monte Carlo quadrature, this next step should be be intuitive:

$$E_V(\boldsymbol{\alpha}) \approx \frac{1}{n}\sum_{i=1}^{n} E_L(\mathbf{X}_i;\boldsymbol{\alpha}) \tag{30}$$

where the n samples $\mathbf{X}_i$ are drawn from the probability distribution $P_L(\mathbf{X};\boldsymbol{\alpha})$ (so in a sense they depend on the parameters $\boldsymbol{\alpha}$). This can be achieved using the Metropolis-Hastings algorithm.

3.4.1 Why Monte Carlo?

There is nothing in the Variational form of Scröndiger's equation that prohibits the use of your favourite technique for the evaluation of the integrals. One could try to evaluate Eq. (?) using e.g., Simpson's rule. When interested in one-particle in 1,2, or 3 dimensions, this might suffice. However, in the case of many-body systems, this would lead one to discover the curse of dimensionality: when using a quadrature rule whose power of the leading error is p, the integration error for a d dimensional integral scales with the number of function evaluations $\mathcal{N}$ we're willing to do as $\mathcal{N}^{-p/d}$. For a 3-dimensional system of N particles, $d = 3N$, and for Simpson's rule $p = 4$ leading to a scaling of $\mathcal{N}^{-1/(0.75N)}$. Then, even for a relatively small many-body system with $N = 10$, doubling the number of function evaluations divides the error by ≈ 1.01.

3.5 Trial wavefunction

The trial wavefunction is our best guess for how the ground-state of Eq. (27) looks like. *One cannot prove within VMC that the guess is exactly correct.* However, due to the variational principle, we can guarantee that the energy calculated with it is an upper bound of the ground-state energy. This creates a practical strategy: one can keep updating the guesses until the calculated energy doesn't change. As we will see later, this can be automatized to an extent. In what follows we'll go over some standard choices for trial-wavefunctions.

A Slater Determinant As already mentioned above a standard choice for the ground-state's form is the Slater Determinant (SD); see Lecture 1. This corresponds to perturbation-theory inspired strategy mentioned above, i.e., splitting into two parts $\hat{H} = \hat{H}_0 + \hat{H}_1$, with $\hat{H}_0 = \hat{K}$ and using eigenstates of $\hat{H}_0$ as trial states:

$$\Phi_0^{(\mathrm{SD})}(\mathbf{X};\mathbf{k}_1,\mathbf{k}_2,\ldots,\mathbf{k}_N) = \mathcal{A}[\phi_{\mathbf{k}_1}(\mathbf{r}_1),\phi_{\mathbf{k}_2}(\mathbf{r}_2),\ldots,\phi_{\mathbf{k}_N}(\mathbf{r}_N)] = \det\begin{pmatrix} \phi_{\mathbf{k}_1}(\mathbf{r}_1) & \phi_{\mathbf{k}_1}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_1}(\mathbf{r}_N) \\ \phi_{\mathbf{k}_2}(\mathbf{r}_1) & \phi_{\mathbf{k}_2}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_2}(\mathbf{r}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{\mathbf{k}_N}(\mathbf{r}_1) & \phi_{\mathbf{k}_N}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_N}(\mathbf{r}_N) \end{pmatrix} \tag{31}$$

where $\phi_{\mathbf{k}_i}(\mathbf{r}_j)$ is an eigenstate of K with with energy $E_i = \hbar^2 k_i^2/2m$ evaluated at the coordinate of particle j:

$$K\phi_{\mathbf{k}_i}(\mathbf{r}) = \frac{\hbar^2}{2m}k_i^2\phi_{\mathbf{k}_i}(\mathbf{r}) \tag{32}$$

In the case of a spin-balanced system of N particles, that is, with $N/2$ particles per species(spin-up and spin-down)[1], we can generalize this trial wavefunction to the product of two SDs:

$$\Psi_0^{(\mathrm{SD})}\left(\mathbf{X};\mathbf{k}_1,\mathbf{k}_2,\ldots,\mathbf{k}_{N/2},\mathbf{k}_{1'},\mathbf{k}_{2'},\ldots,\mathbf{k}_{N/2'}\right)=\Phi_0^{(\mathrm{SD})}\left(\mathbf{k}_1,\mathbf{k}_2,\ldots,\mathbf{k}_{N/2}\right)\Phi_0^{(\mathrm{SD})}\left(\mathbf{k}_{1'},\mathbf{k}_{2'},\ldots,\mathbf{k}_{N/2'}\right) \tag{33}$$

$$=\mathcal{A}[\phi_{\mathbf{k}_{1'}}(\mathbf{r}_{1'}),\phi_{\mathbf{k}_{2'}}(\mathbf{r}_{2'}),\ldots,\phi_{\mathbf{k}_{N'/2}}(\mathbf{r}_{N'/2})]\mathcal{A}[\phi_{\mathbf{k}_1}(\mathbf{r}_1),\phi_{\mathbf{k}_2}(\mathbf{r}_2),\ldots,\phi_{\mathbf{k}_{N/2}}(\mathbf{r}_{N/2})] \tag{34}$$

$$=\det\begin{pmatrix}\phi_{\mathbf{k}_1}(\mathbf{r}_1) & \phi_{\mathbf{k}_1}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_1}(\mathbf{r}_N)\\ \phi_{\mathbf{k}_2}(\mathbf{r}_1) & \phi_{\mathbf{k}_2}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_2}(\mathbf{r}_N)\\ \vdots & \vdots & \ddots & \vdots\\ \phi_{\mathbf{k}_N}(\mathbf{r}_1) & \phi_{\mathbf{k}_N}(\mathbf{r}_2) & \cdots & \phi_{\mathbf{k}_N}(\mathbf{r}_N)\end{pmatrix}\det\begin{pmatrix}\phi_{\mathbf{k}_{1'}}(\mathbf{r}_{1'}) & \phi_{\mathbf{k}_{1'}}(\mathbf{r}_{2'}) & \cdots & \phi_{\mathbf{k}_{1'}}(\mathbf{r}_{N'})\\ \phi_{\mathbf{k}_{2'}}(\mathbf{r}_{1'}) & \phi_{\mathbf{k}_{2'}}(\mathbf{r}_{2'}) & \cdots & \phi_{\mathbf{k}_{2'}}(\mathbf{r}_{N'})\\ \vdots & \vdots & \ddots & \vdots\\ \phi_{\mathbf{k}_{N'}}(\mathbf{r}_{1'}) & \phi_{\mathbf{k}_{N'}}(\mathbf{r}_{2'}) & \cdots & \phi_{\mathbf{k}_{N'}}(\mathbf{r}_{N'})\end{pmatrix} \tag{35}$$

A BCS determinant A state with the lowest rang of many-particle correlations is the BCS state which can be written in the determinant form (see Lecture 1&2)

$$\Psi_0^{(\mathrm{BCS})}(\mathbf{X};\varphi)=\mathcal{A}[\varphi(\mathbf{r}_1,\mathbf{r}_2),\varphi(\mathbf{r}_3,\mathbf{r}_4),\ldots,\varphi(\mathbf{r}_{N-1},\mathbf{r}_N))]=\det\begin{pmatrix}\varphi(\mathbf{r}_1,\mathbf{r}_{1'}) & \varphi(\mathbf{r}_1,\mathbf{r}_{2'}) & \cdots & \varphi(\mathbf{r}_1,\mathbf{r}_{N'/2})\\ \varphi(\mathbf{r}_2,\mathbf{r}_{1'}) & \varphi(\mathbf{r}_2,\mathbf{r}_{2'}) & \cdots & \varphi(\mathbf{r}_2,\mathbf{r}_{N'/2})\\ \vdots & \ddots & \vdots & \\ \varphi(\mathbf{r}_{N/2},\mathbf{r}_{1'}) & \varphi(\mathbf{r}_{N/2},\mathbf{r}_{2'}) & \cdots & \varphi(\mathbf{r}_{N/2},\mathbf{r}_{N'/2})\end{pmatrix} \tag{36}$$

which depends functionally on the *pair wavefunctions*, or *geminals*,

$$\varphi(\mathbf{r}_i,\mathbf{r}_{j'};\{c_n\},t)=\sum_n c_n\phi_{\mathbf{k}_n}(\mathbf{r}_i)\phi_{t(\mathbf{k}_n)}(\mathbf{r}_{j'}) \tag{37}$$

which is in turned defined by the pairing amplitudes c_n and the *pairing scheme* t which pairs single-particle state mfk to state $t(\mathbf{k})$. Note that instead of pairing momentum states one could also pair linear combination of them in the generalized pair wavefunction

$$\tilde{\varphi}(\mathbf{r}_i,\mathbf{r}_{j'};\{c_n\},t)=\sum_n \tilde{c}_n g_n(\mathbf{r}_i)g_{\tilde{t}(n)}(\mathbf{r}_{j'}) \tag{38}$$

where

$$g_n(\mathbf{r})=\sum_m a_{nm}\phi_{\mathbf{k}_m}(\mathbf{r}) \tag{39}$$

This is more similar to the HFB state discussed in Lecture 2.

For example, in the simplest case of the singlet-pairing encountered in terrestial superconductors, the BCS pair state in Eq. (37) is a good phenomenological model with the pairing scheme t corresponding to the time-reversal operator and the pairing amplitudes from BCS theory,

$$t(\mathbf{k})=-\mathbf{k}\ ,\quad c_n=\frac{v(\mathbf{k})}{u(\mathbf{k})} \tag{40}$$

The right choice of trial wavefunction is important for two reasons which persist both in VMC and DMC. First, as we already saw, the trial wavefunction should not be orthogonal to the ground state (or any targeted state), so in a sense it should be a qualitative description of the system, i.e., posses the same symmetries. Second, as we will see later in detail, the trial wavefunction is used to guide the sampling performed in DMC identically to the importance sampling introduced in the Monte Carlo quadrature, afterall the trial wavefunction is our best guess for the ground state of the system. To understand the importance of the second reason we can compare DMC calculations with and without importance sampling: the former vastly outperform the latter and set the limits of what is possible.

[1] In the absence of spin-spin interactions, no spin-flips can occur and it is common to refer to each spin population as "species".

4 Diffusion Monte Carlo (DMC)

In the previous section we provided a robust way to produce an upper bound to the ground state energy. Variational Monte Carlo achieves this by minimizing the variational energy (calculated using Monte Carlo techniques) in a sub-space of the system's Hilbert space, which is parametrized by a set of variational parameters $\boldsymbol{\alpha}$. Depending on the choice of the functional form for the trial wavefunction the minimum of E_V (i.e., the tightest bound), can differ substantially from the ground state energy and approaching further might require substantial insight into the structure of the ground state, which we are looking for in the first place, or a prohibitively large number of variational parameters. In this sense, VMC is not easily systematically improvable.

4.1 Projection via imaginary time-evolution

In this section we will attempt a solution to this problem, that is, identifying the ground state energy, and properties of the ground state, through a clever recasting of Schröndiger's equation. We will describe what will turn out to be Diffusion Monte Carlo, which in principle, is systematically improvable and exact. Diffusion Monte Carlo, relies on the observation that Schröndiger's equation is a diffusion equation in imaginary time. Note that, up to here, we have been concerned with the time-independent Schröndiger's equation and this will remain the case; we're only using the time dependence as a trick to gain insight in the system's ground-state.

We start from the time-dependent Schröndiger's equation for a many-body state $|\Psi\rangle$,

$$i\partial_t |\Psi(t)\rangle = (H - E_T) |\Psi(t)\rangle \ , \tag{41}$$

where E_T is an arbitrary energy offset. Equation 41) has the general solution

$$|\Psi(t)\rangle = U(t) |\Psi(0)\rangle \tag{42}$$

where we defined the time evolution operator

$$U(t) = e^{it(H-E_T)} \ . \tag{43}$$

whose coordinate space representation is a Green's function,

$$G(\mathbf{X} \leftarrow \mathbf{X}', t) = \langle \mathbf{X}| e^{it(H-E_T)} |\mathbf{X}'\rangle \tag{44}$$

which we can integrate with the many-body wavefunction t time t_0 over space to get the wavefunction's value at a time $t' = t_0 + t$:

$$\Psi(\mathbf{X}, t') = \int d\mathbf{X}' G(\mathbf{X} \leftarrow \mathbf{X}', t)\Psi(\mathbf{X}', t_0) \ . \tag{45}$$

We defined this object because we will soon move from the abstract representation with kets to a coordinate space representation which is closer to practical applications. Making the substitution $t = i\tau$, we get

$$-\partial_\tau |\Psi\rangle = H |\Psi\rangle \ , \quad \text{and} \qquad U(t) = e^{-\tau(H-E_T)} = \bar{U}(\tau) \ . \tag{46}$$

That is in imaginary time, Schröndiger's equation becomes a diffusion equation. This means that the imaginary time evolution of any initial state $|\Psi_T\rangle$ will eventually converge on the eigenstate of the Hamiltonian with the lowest energy (which is not orthogonal to the initial state). This is easy to show using an expansion like we did in Eq. (**??**):

$$|\Psi(\tau)\rangle = \bar{U}(\tau) |\Psi_T\rangle = \bar{U}(\tau) \sum_{n=0}^{\infty} a_n |\Phi_n\rangle = \sum_{n=0}^{\infty} a_n e^{-\tau(E_n - E_T)} |\Phi_n\rangle \to a_0 e^{-\tau(E_0-E_T)} |\Phi_0\rangle \tag{47}$$

where we assumed that $E_T \approx E_0$ and that $a_0 = \langle \Phi_0 | \Psi_T \rangle \neq 0$. This property is the basis of DMC. However Eq. (47) is far from how this is implemented in practice. To see that let's first start from the simplest case: the free gas.

The free gas case corresponds to neglecting the interacting terms in the Hamiltonian in Eq. (26). This in turn corresponds to the following Schröndiger's equation:

$$-\partial_\tau |\Psi\rangle = T |\Psi\rangle \ , \quad \text{or in coordinate space} \qquad -\partial_\tau \Psi(\mathbf{X}, \tau) = \frac{1}{2} \sum_{i=1}^{N} \frac{1}{2m} \boldsymbol{\nabla}_i^2 \Psi(\mathbf{X}, \tau) \ . \tag{48}$$

If we interpret Φ as a *density* then this is a $3N$-dimensional diffusion equation. Note that of course Φ is not a density, or a probability density, but rather a probability amplitude, and we need a modulus squared to get a probability density. This is a major issue because it means that unlike any kind of density Φ can become negative when describing fermions and the interpretation of Eq. (48) as a diffusion equation is incomplete. This is the origin of the fermion sign problem. We will ignore it for now, lay the groundwork for DMC and come back to it later where we will see that all of it applies in each pocket of $3N$-dimensional space where Φ has a definite sign. The Green's function of a diffusion equation is

$$G(\mathbf{X} \leftarrow \mathbf{X}', \tau) = (2\pi\tau)^{-\frac{3N}{2}} \exp\left[-\frac{m|\mathbf{X} - \mathbf{X}'|^2}{2\tau} \right] \ . \tag{49}$$

Now we reintroduce the interacting terms in the Hamiltonian, that is we take the entire Hamiltonian from Eq. (26), and the form of the Green's function depends on the form of the two-body, three-body, etc., potentials and in principle does not have a closed form. However, we can write the form of the Green's function for a small time-step τ using the so-called Trotter-Suzuki decomposition which states that

$$\exp[-\tau(A + B)] = \exp[-\tau B/2] \exp[-\tau A] \exp[-\tau B/2] + \mathcal{O}(\tau^3) \ , \tag{50}$$

which is based on the Zassenhaus formula,

$$\exp[\alpha(A + B)] = \exp[\alpha A] \exp[\alpha B] \exp[-\alpha^2 [A, B]/2] \exp[\alpha^3 (2[B, [A, B]] + [A, [A, B]])] \cdots \tag{51}$$

Note that in the Trotter-Suzuki decomposition, the term that is second order in τ cancels. Using this decomposition, ignoring higher-order terms, and taking advantage of the simplicity of V's action on any position state $\mathbf{R}$ we get

$$G(\mathbf{X} \leftarrow \mathbf{X}', \tau) = \langle \mathbf{X} | \, e^{-\tau(T+V-E_T)} \, |\mathbf{X}'\rangle = e^{-\tau[V(\mathbf{X})-E_T]/2} \langle \mathbf{X} | \, e^{-\tau T} \, |\mathbf{X}'\rangle \, e^{-\tau[V(\mathbf{X}')-E_T]/2} \tag{52}$$

$$= (2\pi\tau)^{-\frac{3N}{2}} \exp\left[-\frac{m|\mathbf{X} - \mathbf{X}'|^2}{2\tau} \right] e^{-\tau[V(\mathbf{X})+V(\mathbf{X}')-2E_T]/2} \tag{53}$$

In practice one starts by describing the wavefunction as a collection of M_w walkers at positions $\mathbf{X}_k$, for $k = 1, M_w$, with a density representing the value of the *wavefunction* (*not* the square of the wavefunction), i.e., more walkers where the wavefunctions if larger. The Green's function is then interpreted as a probability density for the evolution of the walkers which then evolve until convergence. Since we only have access to the short-time propagator of Eq. (53), the evolution must be done in very small time-steps τ keeping the error from the Trotter-Suzuki approximation in Eq. (50) (which goes like τ^3 under control. Moreover, the final term in the short-time Green's function in Eq. (53) allows us to intepret the propagation as that of a free system (i.e., with the Green's function of Eq. (49) where the Green's function gets renormalized at every step by a factor $P = \exp[-\tau(V(\mathbf{X}) + V(\mathbf{X}') - 2E_T]$. In the analogy with the diffusion equation, the potential acts as a source/sink of walkers, depending on the sign of the potential (attractive/repulsive). This interpretation allows one to use a birth/death approach where the free Green's function is sampled instead from Eq. (49) and a renormalization is applied by evaluating P and killing a walker with probability $1 - P$. If $1 - P < 0$ a walker is created with probability $P - 1$. Equivalently one can use attach a weight to each walker and accumulate the product.

4.2 Importance sampling

We presented above a simple algorithm for the implementation of DMC. However, in practice this is very inefficient. The efficiency of DMC can be greatly improved by introducing importance sampling which works similarly to the method by the same name in Monte Carlo quadrature: one uses a priori knowledge of the outcome to achieve better sampling.

The underlying idea of importance sampling is that we might have some intuition as to what the system's ground state will look like and we use that as a trial wavefunction. In practice these are low-resolution descriptions of the system, e.g., a mean-field description, or a model that captures the expected structure qualitatively. In general, the trial wavefunction is thought as "almost" the true ground state of the system and we want to find what's missing. Following this logic it makes sense to define a function $f(\mathbf{X},\tau) = \Psi(\mathbf{x},\tau)\Psi_T(\mathbf{X})$, where $\Psi(\mathbf{X},\tau)$ is a wavefunction that satisfies the imaginary time Schröndiger's equation in Eq. (46). The time evolution of $f(\mathbf{X},\tau)$ then is

$$-\partial_\tau f(\mathbf{X},\tau) = -\frac{1}{2}\sum_{i=1}^{N} \boldsymbol{\nabla}^2 f(\mathbf{X},\tau) + \sum_{i=1}^{N} \boldsymbol{\nabla}\cdot\left[\mathbf{v}_D^{(i)} f(\mathbf{X},\tau)\right] + [E_L(\mathbf{X}) - E_T] f(\mathbf{X},\tau) \tag{54}$$

where

$$\mathbf{v}_D^{(i)} = \boldsymbol{\nabla}_i \ln|\Psi_T(\mathbf{X},\tau)| = \frac{\boldsymbol{\nabla}_i\Psi_T(\mathbf{X})}{\Psi_T(\mathbf{X})}\ , \quad E_L(\mathbf{X}) = \frac{H\Psi_T(\mathbf{X})}{\Psi_T(\mathbf{X})}\ . \tag{55}$$

The quantity $\mathbf{v}_D^{(i)}$ is called the *drift velocity* as it represents a velocity terms that carries walkers towards regions where $\Psi_T(\mathbf{X})$ is larger (note that $\mathbf{v}_D^{(i)}$ is proportional to the momentum of particle i in a many-body system described by $\Psi_T(\mathbf{X})$). Since we expect $\Psi(\mathbf{X})$ to be similar to the true ground state this provides better sampling in regions of space that matter. You might also recognize $E_L(\mathbf{X}$ as the local energy introduced for VMC.

The short-time Green's function for Eq. (54) is $\tilde{G}(\mathbf{R} \leftarrow \mathbf{R}',\tau)$ and it is a modified version of $G(\mathbf{R} \leftarrow \mathbf{R}',\tau)$ in Eq. (53):

$$G(\mathbf{X} \leftrightarrow \mathbf{X}') = (2\pi\tau)^{-\frac{3N}{2}} \exp\left[-\frac{m|\mathbf{X} - \mathbf{X}' - 2\tau\mathbf{V}_D(\mathbf{X}')|^2}{2\tau}\right] e^{-\tau[E_L(\mathbf{X}) + E_L(\mathbf{X}') - 2E_T]/2} \tag{56}$$

where $\mathbf{V}_D = \left(\mathbf{v}_D^{(1)}, \mathbf{v}_D^{(2)}, \ldots, \mathbf{v}_D^{(N)}\right)$. This new version of the Green's function has many advantages. As already mentioned, the drift velocity carries the walkers towards regions of increasing $\Psi(\mathbf{X})$, improving the sampling in these regions. Additionally, the renormalization term has now turned to $\tilde{P} = \exp\{-\tau[E_L(\mathbf{X}) + E_L(\mathbf{X}') - 2E_T]/2\}$ which greatly diminishes the fluctuations in populations, if we have chosen a good $\Psi_T(\mathbf{X})$. This is because in that case $E_L(\mathbf{X})$ is close to the ground state energy and roughly constant in space. Finally, the drift velocity makes the fixed-node approximation much more efficient because it pushes walkers away from the nodal surfaces where $\Psi_T(\mathbf{X})$ vanishes by definition.

4.3 The ground state energy and other quantities

The procedure described above results in a set of walkers distributed according to the distribution $f(\mathbf{X},\tau) = \Psi(\mathbf{X},\tau)\Psi_T(\mathbf{X},\tau)$. In this section we will describe how one can use these walkers to calculate various quantities of the many-body system.

For calculating the energy typically the mixed-estimator is used which is given by

$$E_D(\tau) = \frac{\left\langle e^{-\tau H/2}\Psi_T\right| H \left|e^{-\tau H/2}\Psi_T\right\rangle}{\left\langle e^{-\tau H/2}\Psi_T | e^{-\tau H/2}\Psi_T\right\rangle} \tag{57}$$

$$= \frac{\left\langle e^{-\tau H}\Psi_T\right| H \left|\Psi_T\right\rangle}{\left\langle e^{-\tau H}\Psi_T | \Psi_T\right\rangle} \tag{58}$$

For large τ the projector $e^{\tau H}$ projects out of the Ψ_T the ground state and so

$$\lim_{\tau\to\infty} E_D(\tau) = \frac{\langle\Psi_0| H |\Psi_T\rangle}{\langle\Psi_0|\Psi_T\rangle} = \lim_{\tau\to\infty} \frac{\int d\mathbf{X} f(\mathbf{X},\tau) E_L(\mathbf{X})}{\int d\mathbf{X} f(\mathbf{X},\tau)} \tag{59}$$

$$= \frac{1}{M}\sum_{m=1}^{M} E_L(\mathbf{X}_m) \tag{60}$$

Note that since $|\Psi_0\rangle$ is an eigenstate of the Hamiltonian $\langle\Psi_0| H |\Psi_T\rangle / \langle\Psi_0|\Psi_T\rangle = \langle\Psi_0| H |\Psi_0\rangle / \langle\Psi_0|\Psi_0\rangle$, which is called the *pure estimator*. For other quantities that don't commute with the Hamiltonian, the pure estimator is extracted from the *extrapolated estimator*. We will show below how that comes about.

For an operator O with $[H, O] \neq 0$, if we write the ground state $|\Psi_0\rangle$ as $|\delta\Psi\rangle$ away from the trial wavefunction, that is $|\Psi_0\rangle = |\Psi_T\rangle + |\delta\Psi\rangle$, then the pure estimate for O is

$$\begin{aligned}\langle\Psi_0| O |\Psi_0\rangle &= (\langle\Psi_T| + \langle\delta\Psi|)\, O\, (|\Psi_T\rangle + |\delta\Psi\rangle) = 2\langle\Psi_T| O |\Psi_0\rangle - \langle\Psi_T| O |\Psi_T\rangle + \langle\delta\Psi_T| O |\delta\Psi\rangle \\ &\approx 2\langle\Psi_T| O |\Psi_0\rangle - \langle\Psi_T| O |\Psi_T\rangle \end{aligned} \tag{61}$$

where in the last line we have dropped terms quadratic in the (presumably) small $|\delta\Psi\rangle$. The first term in Eq. (61) is twice the mixed estimate while the second is the variational estimate which can be calculated using VMC. You can clearly see that the extrapolated estimate has an error proportional to $(\delta\Psi)^2 = (\Psi_0 - \Psi_T)^2$, that is, it's worse the further the trial wavefunction is from the ground state. As such one typically tries to avoid calculating extrapolated estimates and instead tries to relate quantities to the energy which can be calculated using the mixed estimator (see OES in Lecture 2).